

Project Planning

CrediShield AI — Credit Card Approval Prediction System

1. Project Timeline (Gantt Summary)

The project follows a rapid 12-day development lifecycle, structured into distinct phases from ideation to deployment.

Phase	Task Description	Duration	Start Day	End Day	Status
Phase 1	Brainstorming, Ideation & Problem Definition	1 day	Day 1	Day 1	☑ Completed
Phase 2	Requirement Analysis & Empathy Mapping	1 day	Day 2	Day 2	☑ Completed
Phase 3	Solution Design & Architecture Planning	1 day	Day 3	Day 3	☑ Completed
Phase 4	Dataset Collection & Exploratory Data Analysis	1 day	Day 4	Day 4	☑ Completed
Phase 5	Data Preprocessing & Feature Engineering	1 day	Day 5	Day 5	☑ Completed
Phase 6	Model Training, Evaluation & Selection	1 day	Day 6	Day 6	☑ Completed
Phase 7	Flask Backend Development (<code>app.py</code>)	1 day	Day 7	Day 7	☑ Completed
Phase 8	Frontend UI Development (Glassmorphism & Logic)	2 days	Day 8	Day 9	☑ Completed
Phase 9	System Integration & End-to-End Testing	1 day	Day 10	Day 10	☑ Completed
Phase 10	Cloud Deployment (Render.com) & Troubleshooting	1 day	Day 11	Day 11	☑ Completed
Phase 11	Comprehensive Documentation & Submission Prep	1 day	Day 12	Day 12	☑ Completed

2. Resource Allocation

2.1 Software & Tools

Resource	Category	Purpose
Python 3.12	Language Runtime	Core programming language for ML and Backend
VS Code / Cursor IDE		Code writing, debugging, and terminal access
Git	Version Control	Local versioning and commit management
Jupyter Notebook	Data Science Tool	Interactive EDA, visualizations, and prototyping

2.2 External Platforms & Services

Resource	Category	Purpose	Cost
Kaggle	Data Source	Downloading <code>application_record.csv</code> and <code>credit_record.csv</code>	Free
GitHub	Code Repository	Hosting source code; CI/CD trigger mechanism	Free
Render.com	Cloud Hosting	Hosting the Flask web application in production	Free Tier
Chart.js (CDN)	Frontend Library	Rendering interactive dashboards	Free
Google Fonts	Typography	Supplying 'Plus Jakarta Sans' font	Free

2.3 Human Resources

- **Team Size:** 1 Developer (Solo Project)
- **Roles Assumed:** Product Manager, Data Scientist, Backend Engineer, Frontend Developer, QA Tester, DevOps Engineer, Technical Writer.

3. Risk Management Plan

3.1 Anticipated Risks & Mitigations

Risk Description	Category	Probability	Impact	Mitigation Strategy
Class Imbalance: Model predicts "Approved" for everyone due to 98% safe cases.	Machine Learning	High	High	Use stratified train-test splits; evaluate models using F1-score rather than just accuracy; test Balanced Random Forest.
Library Version Conflicts: Deserialization errors (<code>_loss</code> module missing) on deployment due to <code>scikit-learn</code> version mismatch.	DevOps	Medium	High	Pin exact package versions in <code>requirements.txt</code> to match the local training environment.
Bundle Size Limits: Cloud provider rejects deployment because ML libraries exceed size quotas (e.g., Vercel's 500MB limit).	DevOps	High	High	Exclude training-only libraries (<code>matplotlib</code> , <code>seaborn</code>) from production <code>requirements.txt</code> ; switch from Vercel to Render.
Gunicorn Compatibility: <code>pkg_resources</code> error on Python 3.12.	Deployment	Low	Medium	Upgrade Gunicorn to v22.0.0 which drops the deprecated <code>setuptools</code> dependency.
Large File Commits: GitHub rejects push due to files > 100MB.	Version Control	Medium	Medium	Configure <code>.gitignore</code> early to exclude datasets (<code>data/</code>) and large media files (<code>demo_video.mov</code>).

4. Milestones & Deliverables

Milestone	Deliverable	Acceptance Criteria	Status
M1: Data Readiness	Merged & Cleaned Dataset	Binary labels engineered; missing values handled; outliers removed.	☑ Achieved
M2: ML Pipeline Complete	Serialized ML Artifacts	6 models trained; Gradient Boosting selected (≥95% accuracy); artifacts saved to <code>models/</code> .	☑ Achieved
M3: API Functional	Flask Backend (<code>app.py</code>)	<code>/predict</code> and <code>/metrics</code> endpoints return valid JSON responses for test payloads.	☑ Achieved
M4: UI Integrated	Web Dashboard	Glassmorphism design applied; form submission triggers animated SVG gauge and charts.	☑ Achieved
M5: System Deployed	Live Web Service	Application accessible globally via Render URL; HTTPS enabled; port bound correctly.	☑ Achieved
M6: Project Documented	Documentation Suite	8-phase documentation PDFs created; <code>README.md</code> complete.	☑ Achieved

5. Development Methodology

The project utilized an **Iterative Prototyping** approach within an Agile framework

1. **Prototype Phase:** Used Jupyter Notebooks to rapidly explore data, test feature engineering ideas, and prove that a high-accuracy model could be built.
2. **Refactoring Phase:** Migrated notebook code into a modular Python script (`train.py`) to automate training and artifact generation.
3. **API Phase:** Built the Flask application to wrap the serialized model.
4. **Frontend Phase:** Built the UI iteratively, starting with basic HTML/JS, then adding CSS styling, and finally incorporating animations and charts.
5. **Deployment Iterations:** Pushed code to the cloud, encountered expected infrastructure limitations (e.g., bundle sizes, port bindings), fixed them in code, and redeployed until stable.

Project Name: CrediShield AI

Author: Dinesh (dineshTech-creator)

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